

KisanPilot AI

English Translation – Project Document

This document is the English translation of the uploaded KisanPilot project document, preserving its original structure and terminology.

1. What is KisanPilot AI?

KisanPilot AI is an AI-powered smart farming decision-support platform. It acts as a Personal AI Co-Pilot for a farmer's complete farming journey.

2. What problem does KisanPilot solve?

Farmers need to make decisions about crop health, weather, expenses, market prices, government schemes and farm activities. KisanPilot brings these needs together and provides personalized decision support using farm context and history.

3. KisanPilot USP

KisanPilot AI positions itself as a Personal AI Co-Pilot for a Farmer's Complete Farming Journey. Its key differentiators are farm context, farm memory, personalized AI, AI confidence and human expert escalation.

4. Features

My Farm – Digital Farm Profile: Location, farm area, irrigation, crop, variety, sowing date, expected harvest and soil-test information.

Personal Farm Memory: Sowing, irrigation, fertilizer/input records, photos, disease incidents, expert advice, weather events, expenses and sales history.

Marathi AI Chatbot: Text and voice communication in Marathi using farm profile, crop stage and weather history as context.

Voice Khata: Voice-based accounting. A farmer can speak an entry such as ₹1,500 for labour or ₹2,000 for fertilizer, and AI records category, amount, quantity and date. Existing entries can also be updated.

Farm Economics: Tracks income, expenses, net amount and category-wise/season-wise economics.

Farm Goal Tracking: Tracks the farmer's seasonal net-income target, current progress and remaining amount.

What-if Decision Simulator: Compares scenarios and options, such as whether to grow cotton or soybean or compare selling choices.

5. Advanced Features

Crop Doctor / Disease Detection: Upload a crop photo for possible disease, pest or stress analysis with a confidence score and advisory.

AI Expert Escalation: Low-confidence AI cases can be sent for expert consultation with crop, photo, symptoms and weather history in a structured case summary.

Farm Emergency Mode: Crop problem → Photo → AI Severity → Weather → Farm History → Expert/Nearby Krushi Centre integrated workflow.

Crop Health Journey: Compare crop photos and visual growth changes through a timeline.

Weather Action: Uses weather forecasts and farm context to suggest actions.

Risk Radar: Rain, water stress, disease, heat, cost and insurance-readiness risk indicators.

Next Best Action: Suggests the next priority action and explains why.

6. Market, Schemes & Farm Support

Real Bajar Bhav: Market/mandi prices with source, date/time, minimum/maximum/typical price and trends.

Market Comparison + Net Return: Compares price and estimated transport/relevant selling costs to support selling decisions.

Market Trend: Shows 7-day/30-day price trends; it does not guarantee predictions.

PMFBY + Government Schemes: Provides scheme information, eligibility, documents, dates/deadlines and official-source navigation.

Smart Alerts: Scheme deadlines, insurance-related reminders and crop/farm activity alerts.

Nearby Krushi Kendra / Expert: Finds verified nearby agricultural support details, contacts and directions.

Soil Health Memory: Stores soil-test reports including pH, N, P, K and Organic Carbon for comparison.

One-Tap Farm Report: Combines farm profile, crop health, economics, market sales, disease history, expert advice, weather, schemes and current risks.

Season Scorecard: Summarizes season performance, net return, risk, disease and market performance with personalized learnings.

Farm Twin Replay: Replays the season timeline—sowing, rain, irrigation, disease, expert advice, fertilizer, harvest and sale—to analyze what was learned.

Why / Why Not / What If: Explains recommendations and compares alternatives.

Farm Briefing: A one-button Marathi briefing covering farm, weather, crop, market, risk and actions.

Offline-first Basic Features: Basic farm diary and Voice Khata records can work offline and sync when connectivity returns.

Family Farm Mode: Authorized family members can share farm records.

7. KisanPilot User Flow

1. Farmer Registration / Login → 2. Farm Profile → 3. Crop Season → 4. Marathi Chatbot / Voice → 5. Weather + Farm Memory + Crop Data AI Analysis → 6. Next Best Action / Risk Radar → 7. Photo Upload / Crop Doctor → 8. Low Confidence / Expert Escalation → 9. Voice Khata → 10. Bajar Bhav + Market Comparison + Net Return → 11. PMFBY / Government Scheme Alerts & Guidance → 12. Season End Farm Report + Scorecard + Learnings.

8. Marathi Chatbot

The chatbot uses farm context such as weather, soil, farm history, irrigation and forecasts. Farmers can ask questions about labour, fertilizer and other farming needs. Market answers should show the data source and update time.

9. Bajar Bhav Module

Market data should come from official/reliable sources and show the market name, commodity, available price fields and update time. Stale data should be identified. $\text{Net Return} = \text{Market Price} - \text{Estimated Transport/Relevant Selling Costs}$.

10. PMFBY Module

Provides a basic PMFBY scheme overview based on crop, area, season and farmer profile; eligibility and required documents; dates/deadlines and alerts; and official portal/department source navigation. It should never guarantee an insurance claim. Current scheme rules and dates should be refreshed from official sources.

11. Crop Disease Module – Trust & Safety

Photo quality should be checked for focus and leaf framing. AI results should show confidence/uncertainty. Low-confidence cases should be referred to an expert or nearby agricultural support. AI output should not be presented as a guaranteed final diagnosis or treatment. Verified advice and farm history can be saved.

12. Risk Radar

Rain Risk • Water Stress • Disease Risk • Heat Risk • Cost Risk • Insurance/Scheme Readiness

13. KisanPilot Screens / UI

Home: briefing, Next Best Action, Risk Radar. My Farm: farm profile, memory, crop timeline. KisanPilot: Marathi Chatbot + Voice. Simulator: What-if decisions. Crop Doctor: photo analysis + expert. Market: Bajar Bhav + market comparison + net return. Hisab: Voice Khata + expenses + income + profit. Schemes: PMFBY + government schemes. Help: Expert + Nearby Krushi Kendra.

14. High-Level System Architecture

Farmer App/Web → Authentication & User Profile → Farm Management → AI Orchestration Layer → Marathi NLP/Chatbot + Voice Processing + Vision Model + Recommendation/Decision Engine → Farm Memory/Database → External Data Connectors (weather, market, schemes, maps) → Expert Consultation Module → Reports & Notifications. AI recommendations should use farm context, data freshness and confidence. Market/scheme information should show source and update time.

15. Suggested Technology Stack

Frontend: Flutter / React Native (mobile-first). Backend: FastAPI / Node.js. Database: PostgreSQL. Authentication: Firebase Auth / secure backend authentication. AI/NLP: Marathi-capable LLM + retrieval/grounded knowledge layer. Speech: Speech-to-Text + Text-to-Speech supporting Marathi. Vision: Crop disease image model / transfer learning / suitable vision API. Maps: location-based nearby agriculture support search. Notifications: Firebase Cloud Messaging equivalent. Storage: secure object storage for crop photos/documents. Analytics: farm history, expense aggregation and decision logs.

16. Database Core Entities

Users; Farms; Fields; Crops / Crop Seasons; Farm Activities; Irrigation Records; Input/Fertilizer Records; Expenses; Income / Sales; Market Prices; Crop Photos; Disease Analyses; Expert Consultations; Soil Test Reports; Weather Observations/Alerts; Schemes / PMFBY Information; Notifications; Farm Goals; Decision Simulations; Farm Reports / Season Scorecards.

17. MVP Version

1. My Farm 2. Marathi AI Chatbot 3. Voice Khata 4. Crop Doctor prototype 5. Bajar Bhav module 6. PMFBY + selected government schemes 7. Basic What-if Simulator 8. Farm Economics 9. Basic Risk Dashboard.

18. Phase 2 – Advanced Intelligence

Personal Farm Memory; Next Best Action; Crop Health Timeline; Expert Escalation; Farm Goal Tracking; Why / Why Not / What If; One-Tap Farm Report; Season Scorecard.

19. Phase 3 – Future Features

Farm Twin Replay; Anonymous Local Farm Insights; Farm Briefing; Advanced Benchmarking; Offline-first sync improvements; Advanced predictive models; optional IoT integration (not required for MVP).

20. KisanPilot Differentiation

Personal Farm Memory instead of a generic chatbot; Decision Lab for comparing decisions; Voice Khata for natural Marathi speech accounting; AI Expert Escalation for human verification when uncertain; Next Best Action based on farmer context; Market + Transport + Net Return support for selling decisions; Season Learning for season-to-season insights.

21. Privacy, Trust & Responsible AI

Farmer data should remain private and access-controlled. Location, crop photos and financial records require consent. AI output should communicate certainty/uncertainty. Disease, crop treatment and other high-risk/uncertain cases should be referred to experts. Government-scheme and market information should show source/date/time. Future aggregated insights should not expose individual farmer identity. KisanPilot is AI decision support, not a replacement for the farmer or a qualified expert.

22. Complete Farmer Journey

PLAN → SOW → MONITOR → WEATHER → DETECT → ASK AI / EXPERT → MANAGE COST → CHECK MARKET → HARVEST → SELL → CALCULATE PROFIT → LEARN → NEXT SEASON.

23. Project Presentation Short Pitch

KisanPilot AI is a Personal AI Co-Pilot for farmers. It combines farm profile and history with crop, weather, market, economics, AI support and PMFBY/government-scheme information, along with net-return calculation, What-if Simulator and Next Best Action.

24. Tagline

KisanPilot AI — Personal AI Co-Pilot for a Farmer's Complete Farming Journey.